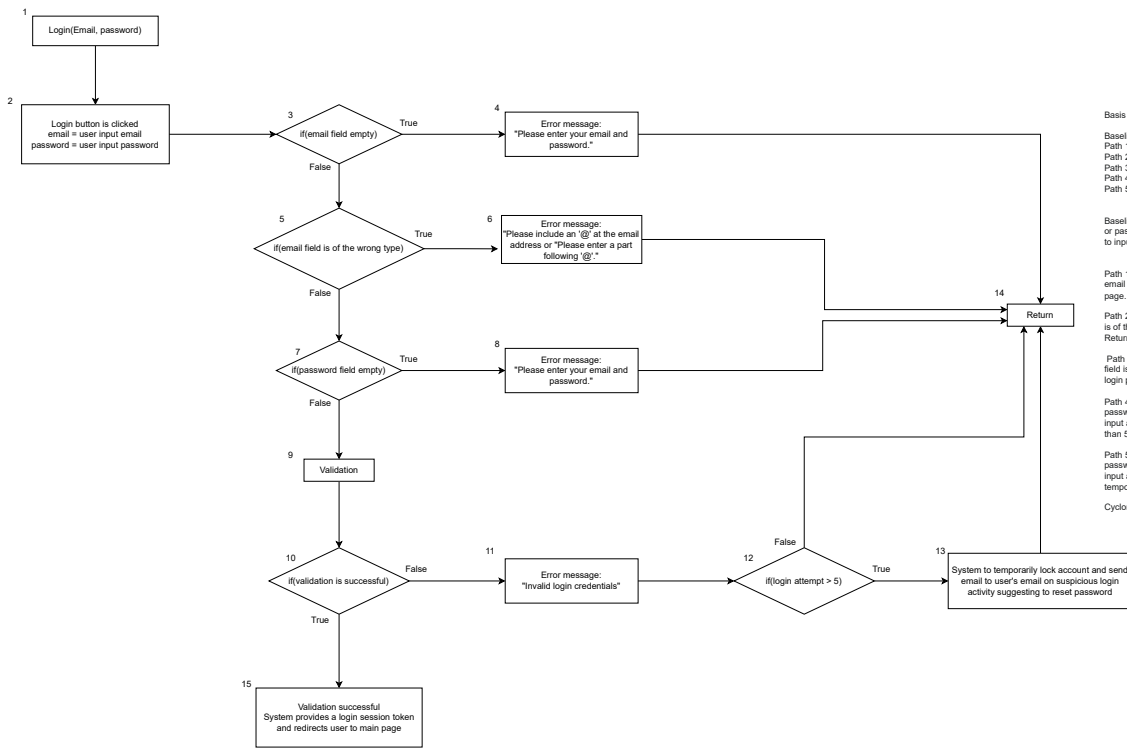




Basis set of path:

Baseline Path: 1, 2, 3, 5, 7, 9, 10, 15

Path 1: 1, 2, 3, 4, 14

Path 2: 1, 2, 3, 5, 6, 14

Path 3: 1, 2, 3, 5, 7, 8, 14

Path 4: 1, 2, 3, 5, 7, 9, 10, 11, 12, 14

Path 5: 1, 2, 3, 5, 7, 9, 10, 11, 12, 13, 14

Baseline Path: User to input login credentials and clicks the login button. System checks for empty email or password fields. No empty fields detected, system to check and validate account details in accordance to input and validation. Validation is successful and brings user to main page.

Path 1: User to input login credentials and clicks the login button. System checks and detects empty email field. System to display error message to prompt user to fill in email field. Returns user to login page.

Path 2: User to input login credentials and clicks the login button. System checks and detects email field is of the wrong type. System to display error message to prompt user to check and re-enter email field. Returns user to login page.

Path 3: User to input login credentials and clicks the login button. System checks and detects password field is empty. System to display error message to prompt user to fill in password field. Returns user to login page.

Path 4: User to input login credentials and clicks the login button. System checks for empty email or password fields. No empty fields detected, system to check and validate account details in accordance to input and validation. Validation is unsuccessful, system to return user to login page if login attempt is less than 5 times.

Path 5: User to input login credentials and clicks the login button. System checks for empty email or password fields. No empty fields detected, system to check and validate account details in accordance to input and validation. Validation is unsuccessful and login attempt is more than 5 times, system to temporarily lock account and email user on suspicious activity suggesting to reset password.

Cyclomatic Complexity V(G)  $5 + 1 = 6$